Advanced Calculus I Non-hand-in Assignment 6

This homework comes from textbook and will **NOT** be collected (這份作業完成後不需要繳交、評分。). But for practice,or quizzes or exams, please make sure that you have idea to solve the following questions by yourself. (這些題目是很重要的練習，小考、期中、期末考會有類似的題目。練習這些題目的過程中所逐漸形成我們對於所學到的知識的"看法"，會在考試中很有幫助。所以，請一定要了解如何解題與解題背後的想法是什麼。)

Please prove the following propositions are true. (請證明下列命題爲眞。)

Section 2.3.
2.24 Remark (on page 69)
2.25 Remark (on page 69)

Section 9.5.
9.32 Theorem (page 325)

第 6 週｜自主練習・課本題目補充

William R. Wade | An Introduction to Analysis | Fourth Edition, Global Edition

依教師 Non-hand-in Assignment 指定順序整理；以下為題目／命題敘述，不含解答。

區間套性質：非空、有界、閉區間形成巢狀區間套時，交集非空。

以下兩則備註要求檢查其中的假設是否必要。

Remark 2.24

Wade p.69

> **2.24 Remark.** *The Nested Interval Property might not hold if "closed" is omitted.*

請構造反例，說明移除「閉區間」假設後，區間套性質可能不成立。

Remark 2.25

Wade p.69

> **2.25 Remark.** *The Nested Interval Property might not hold if "bounded" is omitted.*

請構造反例，說明移除「有界」假設後，區間套性質可能不成立。

Theorem 9.32

Wade p.325

> **9.32 Theorem.** *Let* $\mathbf{Q}^n := \{\mathbf{x} \in \mathbf{R}^n : x_j \in \mathbf{Q} \text{ for } j = 1, 2, \ldots, n\}$. *For each* $\mathbf{a} \in \mathbf{R}^n$ *there is a sequence* $\mathbf{x}_k \in \mathbf{Q}^n$ *such that* $\mathbf{x}_k \to \mathbf{a}$ *as* $k \to \infty$.